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The invasive hornet *Vespa velutina nigrithorax* is more susceptible to an entomopathogenic fungus than two other hymenopteran species, the wasp *Vespula vulgaris* and the bumblebee *Bombus terrestris*

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Abstract Entomopathogenic fungi are commonly used as biological control agents. Recently, a strain of *Metarhizium robertsii* (Metchnikoff) Sorokin (Hypocreales: Clavicipitaceae) collected in France was identified as a good candidate for controlling invasive populations of the yellow-legged hornet, *Vespa velutina nigrithorax* (Buysson, 1905) (Hymenoptera: Vespidae) in Europe. Since *M. robertsii* is a generalist parasite, its usage as bioagent could be damaging for non-targeted species. Here we compared the lethal effects of three concentrations of this *M. robertsii* strain on *V. v. nigrithorax* and two non-target

species commonly found in Europe, the common wasp *Vespula vulgaris* (Linnaeus, 1758) (Hymenoptera: Vespidae) and the buff-tailed bumblebee *Bombus terrestris* (Linnaeus, 1758) (Hymenoptera: Apidae). Exposure to *M. robertsii* spores altered the survival of all three species but at different degrees. Hornets had a consistently lower survival, even under low spore concentrations, when compared to wasps and bumblebees. The low susceptibility of the beneficial hymenopterans is encouraging in the perspectives of using *M. robertsii* as a biocontrol agent against invasive hornets.

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Introduction

The yellow-legged hornet (*Vespa velutina nigrithorax*) (Buysson, 1905) (Hymenoptera: Vespidae) is an invasive social insect species throughout the world, i.e., in Europe (Haxaire et al. 2006), Korea (Kim et al. 2006), Japan (Sakai and Takahashi 2014; Thiéry and Monceau 2024) and, more recently, in North America (Hoebeke et al. 2024). The fast geographic expansion of this invasive hornet poses major concerns to beekeeping activities and the biodiversity (Monceau et al. 2014).

Current methodologies for controlling hornet populations are still very limited and rather inefficient,

as they mostly rely on handmade traps or the use of pesticides, which could be harmful to a wide range of non-target species (Kishi and Goka 2017; Goulson 2019). Recently, however, several alternative biocontrol options have been explored (Turchi and Derijard 2018) including the use of the entomopathogenic fungus *Metarhizium robertsii* (Metchnikoff) Sorokin (Hypocreales: Clavicipitaceae) (Poidatz et al. 2018). This parasite pertains to the species complex *M. anisopliae* naturally found in the soil (Meyling and Eilenberg 2007). It is already commercialized for biocontrol and known to be efficient against many Orthopteran, Isopteran and Thysanopteran pest species (i.e., *M. anisopliae* strain F52 approved in Europe, USA, Brazil, Columbia and China for the use against various insects; *M. anisopliae* strain FI-985 approved in Australia, India and China for the use against grasshoppers and locusts). Recently, Poidatz et al. (2018) showed that exposure of yellow-legged hornets to spores of four isolates of *M. robertsii* (EF2.5(1), EF3.5(1), EF3.5(2), EF3.5(4)) at 10^7 spores per milliliter (sp ml^{-1}) reduced their survival by half after only five to six days, thereby positioning this fungus as a serious candidate bioagent for hornet population control.

From a biocontrol perspective, entomopathogenic fungi like *M. robertsii*, present the advantage that they can be quickly and broadly transmitted across social host, such as in a hornet nest. Indeed, sociality favors parasite transmission between genetically close individuals that are spatially aggregated, and engage in numerous interactions, through social contacts, food exchange, or proximity with nest materials (Dimbi et al. 2013; Khun et al. 2021). However, the potential infection of non-target species should also carefully be taken into consideration if, for instance, infected hornets leave their nest and disseminate spores from their body surface (Smagghe et al. 2013). In particular, there is a risk that wild and managed pollinators get contaminated through direct contact with spores when contaminated hornets hunt unsuccessfully and release their prey (Ueno 2015), or collect nectar on flowers (Kapongo et al. 2008).

Several studies on the species complex *M. anisopliae* (recently recognized as a separate species within the *M. anisopliae* species complex) (Bischoff et al. 2009; Rehner and Kepler 2017) have reported negligible (Smagghe et al. 2013) to strong negative impact (Shaw et al. 2002) on several

beneficial insects such as honey bees and bumblebees. These impacts were assessed using different strains of *M. anisopliae* (Shaw et al. 2002), and since the recent differentiation of *M. robertsii* as species (Bischoff et al. 2009), determining which species was used based on the new classification is challenging. Moreover, the results of these studies varied depending on the doses used for parasite exposure (Kanga et al. 2003; Smagghe et al. 2013), the formulations (Ball et al. 1994), and the methods of application employed (Rodríguez et al. 2009). For instance, contact with a powder concentrate of 10^7 spores per gram (sp g^{-1}) decreased the survival of bumblebees by less than 25% within six weeks (Smagghe et al. 2013). By comparison, this decrease of survival reached 100% at a concentration of 10^9 sp g^{-1} within the same period of time. The survival of honey bees was lower when the fungus was sprinkled on and between the frames of a hive than when conidia were stamped on filter papers and placed on half the frames (Rodríguez et al. 2009). Given these potential advantages and disadvantages of *M. robertsii* for controlling hornet populations, there is a need to better characterize the impact of exposure with comparable methodologies both on invasive hornets and non-target species before it can be further considered for biocontrol.

Here we compared the lethal effect of a *M. robertsii* strain EF3.5(1) on *V. v. nigrithorax* and two non-target species of wild pollinators commonly found in Western Europe: the common wasp *Vespa vulgaris* (Linnaeus) (Hymenoptera: Vespidae) and the buff-tailed bumblebee *Bombus terrestris* (Linnaeus) (Hymenoptera: Apidae). We hypothesized that *M. robertsii* would have different effects on insects, depending on the ecology and the natural history of the host species. Specifically, we expected that the virulence of the parasite would be stronger in hornets than in wasps and bumblebees, because the non-target species may have evolved some resistance to *M. robertsii* through frequent exposure in their below-ground nests, whereas hornets may be less often exposed in their aerial nests. We tested this hypothesis by exposing wild hornets, wasps, and bumblebees to different concentrations of *M. robertsii* and monitoring their survival for eight consecutive days. We decided to focus on wild and non-target pollinators (wasps and bumblebees)

Table 1 Details about the insect colonies used

	Hornet	Wasp	Bumblebee
Status in Europe	Invasive	Non-invasive	Non-invasive
Nest	Aerial	Underground Occasionally aerial	Underground
Origin	Wild	Wild	Commercial
Number of nests	4	1	6
Country of origin (City, if available)	France (Laburgade) France (Bouliac) France (Bouliac) France (Cahors)	France (Saucats)	Netherlands Netherlands Netherlands Netherlands Netherlands Netherlands
Collection date (year-month-day)	2021-10-12 2021-10-18 2021-10-25 2021-11-28	2021-12-10	2023-11-08 2023-11-08 2023-11-08 2023-11-08 2023-11-08 2023-11-08
Nest diameter at the widest part (cm)	<30 30-80 30-80 >80	60-70	<20 <20 <20 <20 <20 <20
Year of experiments	2021	2021	2023
Number of microcolonies per treatment			
<i>Control</i>	9	6	6
<i>Low concentration</i>	9		6
<i>High concentration</i>	9	6	6

For each species (hornet: *V. v. nigrithorax*, wasp: *V. vulgaris*, bumblebee: *B. terrestris*) we built artificial microcolonies using workers sourced in mother colonies. These individuals were either exposed to distilled water (*control*), 10^3 spores of *M. robertsii* per milliliter (10^3 sp ml⁻¹: *low concentration*), or 10^7 spores of *M. robertsii* per milliliter (10^7 sp ml⁻¹: *high concentration*)

in order to explore potential effects on the wild biodiversity.

Materials and methods

Insects

All the experiments were performed at the INRAe Center of Villenave d'Ornon (France) during summers 2021 and 2023. We used four wild *V. v. nigrithorax* nests, one wild *V. vulgaris* nest, and

four commercial *B. terrestris* nests (see details in Table 1). Although commercially reared since about 30 years, these bumblebee populations originate from wild populations sampled in Europe and are frequently mixed with freshly caught wild individuals (Velthuis and Doorn 2005). Prior to the experiments, we maintained the bumblebee colonies during two weeks in a rearing room (temperature 26 °C, 60% RH) with *ad libitum* access to a commercial mixture of pollen (Icko-apiculture company, France) and 30% (v/v) sugar solution. During this acclimation period, we removed the commercial sugar water initially

available in the colonies in order to avoid any potential contact between the bumblebees and antifungal components. Regarding hornets and wasps, we cooled the nests in a cold chamber (4 °C) immediately after we captured them in order to collect the individuals. Here, an acclimation period was not needed since these wild insects were not exposed to antifungal components. We therefore directly proceeded with the infection as described below.

For each insect species, we collected adult workers from the different nests and grouped them into microcolonies of ten nestmates in clear plastic boxes (72×26.5×25 cm). We selected the individuals pseudo-randomly by targeting all available workers of intermediate body size (avoiding very small and very large individuals) and without any lighter pilosity (indicating young adults that recently emerged from their pupae). At a minimum, we created one microcolony for each treatment from each nest. Then, priority was given to ensure a sufficient number of microcolonies per treatment (minimum of four). In the case of *V. vulgaris*, as we only found one nest, we only tested the control and high concentration treatments (see below). Every microcolony from the same nest was formed at the same time. We kept the microcolonies in incubators (Memmert, Germany) at 26 °C, with around 60% RH and in the dark for the full duration of the experiment. Each microcolony was provided sucrose solution (40% v/v) and water *ad libitum*.

Entomopathogen fungus

We used the strain EF3.5(1) of *M. robertsii* sampled in 2015 in an experimental INRAE vineyard (Villenave d'Ornon, France) (see Poidatz et al. 2018 for details). We selected this isolate because of its high virulence against hornets as shown by direct inoculation and inoculation by transfer between hornets (Poidatz et al. 2018). *M. robertsii* was cultivated on a Petri dish (9 cm diameter) containing oat agar media (Oat flour 40 g and Potatoes Dextrose Agar (Sigma Aldrich), Chloramphenicol (Sigma Aldrich), and permuted water) in a dark room. The fungus was transplanted every 2–3 months to ensure its healthy growth through nutrient resupplying.

Infections

We prepared spore solutions by mixing *M. robertsii* spores (extracted from culture media), distilled water, and 0.02% Tween80 (Sigma Aldrich) (Poidatz et al. 2018; Ponchon et al. 2022). The concentration of the solution was adjusted at 10^3 sp ml⁻¹ (identified in a preliminary experiment as the sublethal concentration for bumblebees, named 'low concentration') or 10^7 sp ml⁻¹ (named 'high concentration') by counting spores in 10 µl using a Neubauer cell. The control treatment was composed of distilled water and 0.02% Tween80 (named 'control'). Each insect was cooled in a tube placed in ice for a few minutes until anaesthetized before being exposed to one of the treatments by immersion of their entire body in the solution for 1 s. We used *M. robertsii* exposure by immersion because of its higher efficacy on hornet survival compared to exposure by contact (infected support or individual) or by food (Poidatz et al. 2018).

Survival assays

We started the survival survey immediately after exposure to spores and the formation of microcolonies. From then on, we recorded the number of daily dead insects and removed them from each microcolony for eight consecutive days.

Statistical analysis

We ran all the analyses in R 4.3.3 (R Core Team 2020). We measured the combined effects of host species and *M. robertsii* spore concentrations on host survival using a proportional hazards regression model (coxph function, survival package, Therneau 2023) with an interaction between the fixed factors 'species' (using *V. v. nigrithorax* as reference) and 'treatment' (using the control as reference). In this analysis, the reported z-values correspond to Wald tests performed by the coxph function (Therneau 2023). To assess the global interaction term, given by the χ^2 , we further performed a likelihood ratio test using the anova function. We calculated an estimate of the survival curve for each treatment within species using the survfit function (survival package, Therneau 2023) with the treatment as a fixed factor.

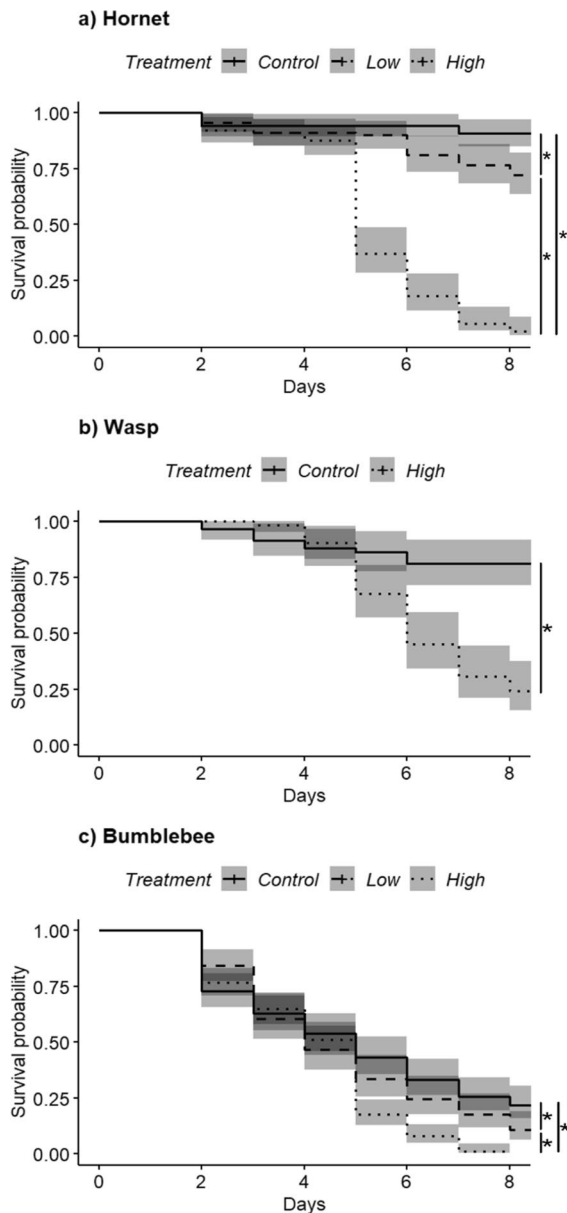


Fig. 1 Survival probability of the three Hymenoptera species over eight days: **a** Hornet (*V. v. nigrithorax*), **b** Wasp (*V. vulgaris*), **c** Bumblebee (*B. terrestris*). We exposed the insects to three treatments: control (continuous line), low concentration of spores (10^3 sp ml^{-1} , dashed line), high concentration of spores (10^7 sp ml^{-1} , dotted line). Solid lines represent experimental data and grey areas the 95% confidence interval. Stars represent significant differences between groups calculated using proportional hazards regression model (hornet: high vs. control: $p < 0.001$, high vs. low: $p < 0.001$, low vs. control: $p = 0.02$; wasp: high vs. low: $p < 0.001$; bumblebee: high vs. control: $p < 0.001$, high vs. low: $p = 0.002$, low vs. control: $p < 0.001$)

Results

Hornets were the most impacted by spore exposure

We first tested the influence of *M. robertsii* concentrations on the survival of hosts (Fig. 1). For all three insect species, individuals exposed to high concentrations of spores showed reduced survival compared to unexposed controls of the same species (see Table 2). Highly-exposed hornets showed a decreased risk in survival by 22.99 times (Cox model, hazard ratio (HR)=22.99, 95% Confidence Interval CI 11.46–46.04, $z = 8.85$, $p < 0.001$) compared to controls. This reduction observed in hornets was much higher than the reduction by 6.22 times (HR=6.22, 95% CI 1.43–27.05, $z = 2.44$, $p < 0.001$) observed in wasps or by 3.84 times (HR=3.84, 95% CI 1.06–13.91, $z = 2.05$, $p < 0.001$) in bumblebees at the same spore concentration. Highly-exposed hornets and bumblebees also showed a decreased survival by 8.71 and 1.78 times, respectively, compared to their lowly-exposed conspecifics (hornets: HR=8.71, 95% CI 5.54–13.68, $z = 9.39$, $p < 0.001$; bumblebees: HR=1.78, 95% CI 1.22–2.60, $z = 3.00$, $p = 0.002$). Finally, lowly exposed hornets and bumblebees showed a decrease in survival compared to controls, by 3.17 and 2.15 times respectively (hornets: HR=3.17, 95% CI 1.49–6.74, $z = 3.01$, $p = 0.002$; bumblebees: HR=2.15, 95% CI 0.553–8.39, $z = 4.36$, $p < 0.001$). The reduction of survival thus varies with the dose of spore exposure. Overall, the difference in survival between exposure doses was significantly larger in hornets than in wasps and in bumblebees (species $\times$ treatment, $\chi^2 = 37.37$, $\text{df} = 3$, $p < 0.001$). This indicates that non-target species (wasps and bumblebees) were less susceptible to spore exposure than hornets.

Hornets and wasps lived longer than bumblebees at all spore concentrations

We then compared the effects of the parasite on host survival across species and for each treatment. In unexposed controls, the hornets and the wasps had similar lifespan (HR=1.981, 95% CI 0.82–4.78, $z = 1.52$, $p = 0.128$) and a higher probability of survival than the bumblebees (hornets vs. bumblebees: HR=9.578, 95% CI 4.62–19.85, $z = 8.21$, $p < 0.001$; wasps vs. bumblebees: HR=4.83, 95% CI 2.47–9.48,

Table 2 Results of the statistical models comparing the survival probability across treatment and species using a Cox model: species: Hornet (reference, *V. v. nigrithorax*), wasp (*V.**vulgaris*), bumblebee (*B. terrestris*); treatments: control (reference, control), low (low spore concentration), high (high spore concentration)

	<i>Coef</i>	<i>HR</i>	<i>95% CI</i>	<i>pval</i>
<i>Species</i>				
<i>SpeciesWasp</i>	0.68	1.98	0.82-4.78	0.128
<i>SpeciesBumblebee</i>	2.26	9.58	4.62-19.85	<0.001
<i>Treatment</i>				
<i>TreatmentLow</i>	1.15	3.17	1.49-6.74	0.003
<i>TreatmentHigh</i>	3.14	22.99	11.48-46.06	<0.001
<i>Species x Treatment</i>				
<i>SpeciesBumblebee x TreatmentLow</i>	-0.39	0.68	0.29-1.61	0.381
<i>SpeciesWasp x TreatmentHigh</i>	-1.31	0.27	0.10-0.70	0.007
<i>SpeciesBumblebee x TreatmentHigh</i>	-1.79	0.17	0.07-0.37	<0.001

Coef (Coefficient): the slope of the regression model, HR (Hazard Ratio): the exponential of the coefficient, which represents the increased risk of an event occurring (in this case, the death of an individual) when comparing one factors variable to the baseline, 95% CI (Confidence Interval): HR 95% confidence interval, and pval (p-value)

$z=4.59$, $p<0.001$). When they were exposed to the parasite at low or high concentrations, the hornets and the wasps lived longer than the bumblebees (low concentrations: hornets vs. bumblebees: HR=6.51, 95% CI 4.08–10.39, $z=7.87$, $p<0.001$; high concentration: hornets vs. bumblebees: HR=1.60, 95% CI 1.15–2.23, $z=2.78$, $p=0.005$; wasps vs. bumblebees: HR=2.98, 95% CI 2.04–4.37, $z=5.61$, $p<0.001$). At high concentrations, the wasps survived longer than the hornets (HR=0.54, 95% CI 0.38–0.76, $z=-3.48$, $p<0.001$). This indicates that bumblebees had the lowest survival probability in this experimental setup compared to hornets and other bumblebees.

Discussion

We exposed workers from three social Hymenoptera species found in Western Europe to different concentrations of a strain of an entomopathogenic fungus recently reported to affect invasive yellow-legged hornets (Poidatz et al. 2018). Overall, for all spore concentrations tested, hornets were more susceptible to the parasite than wasps and bumblebees, comforting the idea that *M. robertsii* is a good candidate for biocontrol against yellow-legged hornets with comparatively smaller impact on non-target species.

Our parasite strain EF3.5(1) harmed differently the survival of the three insect species depending on the exposure concentration. In particular, bumblebees had a shorter lifespan than the two other species

but seemed less sensitive to entomopathogenic concentrations than the hornets. Since *B. terrestris* nests are found underground, it is possible that they have already been in contact with closely related strains of *M. robertsii* during their evolution. Indeed, cases of natural exposure of bumblebees to *Metarhizium* spp. have already been reported (Macfarlane et al. 1995; Schmid-Hempel 2001). In the context of host-parasite coevolution, bumblebees could have developed greater resistance to this parasite (Bérénos et al. 2009), which should less (if at all) be the case for yellow-legged hornets (invasive species, aerial niche) and wasps (local species but subterranean, or aerial niche). Our observations that bumblebees are less sensitive to the parasite support this hypothesis of an evolutionary adaptation. It must be noted, however, that the fact that we observed a low survival of bumblebees in our experiments could be explained by the health status of the source colonies we worked with. It would be interesting to replicate these experiments on wild bumblebees although that, if this bias had been present, it would have influenced our results in the opposite direction (i.e., a greater susceptibility to the parasite).

Interestingly, the effects of *M. robertsii* were the strongest on the yellow-legged hornets. Our observation that only half of the hornets survived four days after the exposition to spores is comparable to the results of Poidatz et al. (2018), thereby confirming the efficiency of *M. robertsii* against hornets. From a biocontrol perspective, the application method

of *M. robertsii* seems key to enhancing selectivity: the direct application of spores into hornets' nests could be interesting as it allows a selective application and the reduction of potential non-target species contamination. Moreover, it is possible that *M. robertsii* causes a range of early post-infection sublethal effects that impair hornet activity before they die. Indeed, many parasites and pathogens can alter the behavior of insects (Gómez-Moracho et al. 2017). *M. anisopliae* has been showed to impact the reproduction of cockroaches (*Blatella germanica*) (Quezada-Moraga et al. 2004), the nutrition of fruit flies (*Drosophila melanogaster*) (Toledo-Hernández et al. 2018) and also the locomotion activity of termites (Hassan et al. 2021). It is, for instance, possible that contaminated hornets are less able to leave their nest and forage, thus greatly limiting the opportunities of contaminating non-target species through contact with water, flowers, or honey bee hives. However, this scenario remains to be tested, and although wild non-target species appear to be less susceptible to spores than hornets, potential adverse effects in the case of intraspecific contamination cannot be ruled out. It would therefore be valuable to complement this study with mortality data on *V. vulgaris* exposed to low spore doses and to screen this and lower doses across a broader range of wild and non-target species to assess the risks of unintended contamination and determine the threshold dose with no impact on survival.

The sublethal effects of *M. robertsii* on hornets, their ability to spread spores in the environment, and the transmission of spores to other pollinators through contaminated surfaces should, therefore, be further considered in order to establish the safety of this biocontrol agent. Further studies on the dynamics of parasite transmission between nestmates will also be necessary to assess the actual quantity of spores required to inject into a hornet nest. Many social insects use behavioral strategies such as grooming or reduced contact with infected individuals in order to limit the transmission of parasites or pathogens in the colony and their establishment in the nest (Cremer et al. 2007). Preliminary experiments done by our group suggest that yellow-legged hornets increase the duration of their grooming behavior when infected by *M. robertsii*. Such hygienic behaviors and potential others could be restrictive in a context of entomopathogen usage.

Our study suggests entomopathogenic fungi, and particularly *M. robertsii* strains, are good candidates for biocontrol of invasive yellow-legged hornets as they showed higher mortality compared to the two non-target species. However, as the effect on non-target species is concentration-dependent, there is a need to further quantify the real risk of spores spreading in the environment by an infected individual to better assess the safety of this biological control agent on local biodiversity.

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Author contributions MLa, FV, MLI and DT designed the study. MLa, NR, BM-C performed the experiments. MLa wrote the first draft. MLa, FV, MLI and DT revised the manuscript.

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Declarations

Conflict of interests The authors declare no competing interests. M2i Biocontrol did not influence the analysis and the conclusions.

Ethical approval The authors declare no competing interests. Guidelines on ethical issues concerning animals were considered and complied with.

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